

AFSS Backlog Tracker — Project Documentation

Version: 1.0
Deployed at: <https://bryan-technical-afss-712513641417.australia-southeast1.run.app>
GCP Project: buoyant-purpose-475203-t9
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1. Overview

The AFSS Backlog Tracker is an internal web dashboard that connects to SimPRO to give management a live, consolidated view of:

- **Backlog** — all CFSP audit jobs across three companies (RM AFSS, CHUBB/AFAC, AE Evac), broken down by status
- **Team supply** — remaining working hours per auditor for the current month, factoring in leave
- **Technical Support Works** — non-audit billable work, invested time, and quality assurance activity scheduled to the tech team

The app is hosted on Google Cloud Run (Sydney region) and updates automatically every 30 minutes via a Cloud Scheduler job.

2. Pages & Features

Backlog Page (/)

The main jobs list. Displays all pending and in-progress CFSP audit jobs across all three companies in a filterable, sortable table.

- Filter by company, stage, status
- Click any row to open the job in SimPRO directly
- Shows job name, customer, site, due date, estimated hours, invoice amount, scheduled date, and assigned technicians

Dashboard (/dashboard)

A one-page management summary with four sections:

1. **Tech Team Works table** — job counts, hours, and invoice amounts grouped by status row and company column
2. **Status rows** — Scheduled Awaiting to be Done / Awaiting Client Info / Tentative Awaiting Scheduling / Attendance Complete-Results To Be Released
3. **Technical Team Supply** — per-auditor hours remaining this month
4. **Technical Support Works** — other billable work, invested time, and QA jobs

3. Dashboard — Section-by-Section

3.1 Tech Team Works Table

Column	Description
Count of Job	Number of CFSP jobs in that status for that company
Sum of Est. Hours	Total estimated labour hours
Sum of Amounts	Total invoice value (ex-tax, or \$330 default if blank)

Total Backlog as at end of period — sums all jobs across all status rows and all companies.

Month filter (top-right) — filters the table to show only jobs whose scheduled date falls in the selected month (also includes the prior month). Defaults to "All".

Status Rows

Row	Colour	Condition
Scheduled Awaiting to be Done	Green	Job has a confirmed scheduled date in SimPRO
Awaiting Client Info	Blue	Job Status Name contains "client" or "awaiting"
Tentative Awaiting Scheduling	Orange	Pending job with no scheduled date confirmed
Attendance Complete/Results To Be Released	Grey	Stage=Progress (all companies except AE Evac), or Status contains "complete/released/attendance"

Note for AE Evac (company 10): Progress-stage jobs are treated as "Scheduled Awaiting to be Done" if they have a date, or "Tentative" if they do not — because AE Evac uses Progress stage to mean "booked", not "done".

Loading States

- While data is loading, **Scheduled** and **Tentative** rows show — (dashes) because the breakdown between the two cannot be determined until full schedule enrichment is complete.
- The amber "**Refreshing schedule data...**" indicator means the background enrichment is still running. Numbers resolve automatically within 1–2 minutes on a cold cache.
- On a warm cache (normal operation), the dashboard loads with correct numbers in under 2 seconds.

3.2 Technical Team Supply

Shows remaining billable supply hours for each of the three AFSS auditors for the **current calendar month**.

Column	Description
--------	-------------

May (current month)	Hours left this month = remaining working days × 8 hrs
Total Supply Hours	Same as the monthly figure (matches the SimPRO "end of period" concept)
Roles	Each auditor's primary function

Leave deduction: If an auditor has confirmed leave booked in SimPRO this month, those days are subtracted from their supply hours. The note at the bottom of this section lists any leave on record for today or upcoming.

Remaining working days logic:

- Before 3:00 PM today → today counts
- From 3:00 PM onwards → today is excluded (work day is done)

3.3 Technical Support Works

Three columns of data, all sourced automatically from SimPRO:

Other Billable Work Scheduled to Tech Teams (100/hr)

- **What:** Jobs scheduled to Josh Roger, Muhammad Soban, Ryan Gordon (and their Tentative placeholders) **excluding** CFSP audit jobs and "system testing" tagged jobs
- **Date range:** Today → end of current month (rolling daily)
- **How:** Scans the SimPRO schedule day-by-day, collects all job-type blocks for those staff members

Invested Time (100/hr)

- **What:** Pending jobs belonging to customers who have work scheduled to the tech team during the same period
- **Date range:** Today → end of current month
- **How:** From schedule blocks, extracts the customer IDs of scheduled jobs, then finds all pending SimPRO jobs for those customers

Quality Assurance — Overall Total of Jobs in SimPRO (100/hr)

- **What:** All Pending + Progress company 1 jobs assigned to "A Quality Assurance Officer"
- **Date range:** Not date-filtered — shows the full current QA workload
- **How:** Fetches all jobs from company 1 filtered to that technician ID

4. Data Sources & Business Logic

SimPRO Connection

All data comes from the SimPRO REST API v1.0 using a Bearer token. The base URL and token are stored as Cloud Run environment variables.

CFSP Filter

A job qualifies as a CFSP audit job if it has technician ID **1126** ("A CFSP ONLY") assigned to it. This filter applies to companies 1 and 10. Company 8 (CHUBB/AFAC) returns all jobs without filtering.

Job Hour Estimation

```
Estimated hours = job.Totals.ResourcesCost.LaborHours.Estimate
If that is 0 or missing → default to 2 hours
```

Job Price

```
Invoice amount = job.Total.ExTax
If that is 0 or missing → default to $330
```

Scheduled Date Source

- A job is considered "scheduled" if any of these exist:
- 1. A schedule block in the SimPRO schedules API for that job (preferred — most accurate)
 - 2. `Scheduled` , `DateScheduled` , `ScheduledDate` , or `DateBooked` field in the job detail

5. API Routes

Route	Purpose	Cache TTL
GET /api/data?company=N&stage=S	Job list for one company/stage combo	1 hour
GET /api/data?company=N&stage=S&force=1	Force refresh (background; returns partial immediately)	—
GET /api/leave	Team leave schedule for current month	1 hour
GET /api/leave?force=1	Force leave refresh	—
GET /api/tech-support	Technical Support Works stats	1 hour
GET /api/tech-support?force=1	Force tech-support refresh	—
GET /api/warmup	Pre-warms all caches (called by Cloud Scheduler)	—

/api/data response fields (per job)

Field	Source	Description
_site	Sites API	Full site object
_scheduledDate	Schedules API	ISO date string of first confirmed schedule block
_scheduledHours	Schedules API	Total scheduled hours across all blocks
_customerGroup	Customers API	Customer group name (e.g. "Government")
_company	Added by dashboard	Company ID (1, 8, or 10)

X-Partial: 1 header

When the response contains jobs that have **not yet been enriched** with schedule, site, and customer data, the API returns this header. The dashboard detects it and automatically retries every 5 seconds until the header is absent.

6. Caching Strategy

Storage

Caches are written to JSON files in the OS temp directory (`/tmp` on Cloud Run). Each company/stage combination has its own cache file.

TTL

All caches expire after **1 hour**.

Stale-while-revalidate

When a request arrives for stale or partial data:

1. The cached data is returned **immediately** (fast response to user)
2. A background refresh kicks off against SimPRO
3. The next request will get the fresh data

Cold cache (no file at all)

1. The fast-path runs: job list is fetched from SimPRO and returned with `X-Partial: 1`
2. Background enrichment starts (fetching per-job details, schedules, sites, customers)
3. Enrichment takes 30–160 seconds depending on job count
4. Dashboard auto-retries every 5 seconds until enrichment completes

Warmup

The `/api/warmup` endpoint runs all three cache types sequentially (job data) and in parallel (leave + tech-support). It is called:

- **Automatically** every 30 minutes by Cloud Scheduler
- **Manually** after any new deployment (by running the warmup PowerShell command)

7. Team Members

Name	SimPRO ID	Role
Muhammad Soban	1581	Primary APFS
Ryan Gordon	15	Mixed of Technical Support Works and Secondary APFS
Josh Roger	1753	Estimation / Office Management Time ETC

Leave references tracked: **1** = Annual Leave, **2** = Sick/Personal Leave.

8. Companies & Job Classification

Company ID	Name	SimPRO Company	Filter
1	RM AFSS	Company 1	CFSP tech ID 1126
8	CHUBB/AFAC AFSS	Company 8	All jobs (no filter)
10	AE Evac Procedure Audits	Company 10	CFSP tech ID 1126

AE Evac rate: \$385/hour (shown in column header). All other companies: \$330 default or actual invoice amount.

Stages fetched: Pending and Progress for each company (6 total combinations).

9. Deployment

Platform

Google Cloud Run, region: australia-southeast1

Service name: bryan-technical-afss

Project: buoyant-purpose-475203-t9

Deploy command

```
gcloud run deploy bryan-technical-afss `
  --source . `
  --region australia-southeast1 `
  --project buoyant-purpose-475203-t9 `
  --quiet
```

After every deployment — warm the cache

```
Invoke-WebRequest `
  -Uri "https://bryan-technical-afss-712513641417.australia-southeast1.run.app/api/warmup" `
  -TimeoutSec 310
```

This takes 3–5 minutes. All six company/stage caches will be populated before any user visits the dashboard.

Build

The app is containerised via the Dockerfile at the project root. Cloud Run builds the container from source using Google Cloud Build.

10. Automated Cache Warming

A Cloud Scheduler job (afss-warmup-job) hits the warmup endpoint every 30 minutes:

- **Schedule:** */30 * * * * (every half-hour, UTC)
- **Region:** australia-southeast1
- **Timeout:** 310 seconds
- **URL:** https://bryan-technical-afss-712513641417.australia-southeast1.run.app/api/warmup

This ensures that even if the Cloud Run instance is recycled (which wipes /tmp), the cache is always refreshed within 30 minutes, long before the 1-hour TTL expires. Under normal operation, users loading the dashboard will always get pre-cached data and see results in under 2 seconds.

11. Environment Variables

Set on the Cloud Run service (not in code):

Variable	Description
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SIMPRO_BASE_URL	SimPRO instance base URL (e.g. <code>https://example.simpro.co</code>)
SIMPRO_TOKEN	SimPRO API Bearer token

12. Tech Stack

Layer	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript 5
UI	React 19, Tailwind CSS v4
Hosting	Google Cloud Run (australia-southeast1)
Scheduling	Google Cloud Scheduler
Data source	SimPRO REST API v1.0
Cache storage	OS temp directory (/tmp JSON files)
Build	Google Cloud Build (from Dockerfile)

Documentation generated May 2026. For issues or questions, contact the system administrator.